

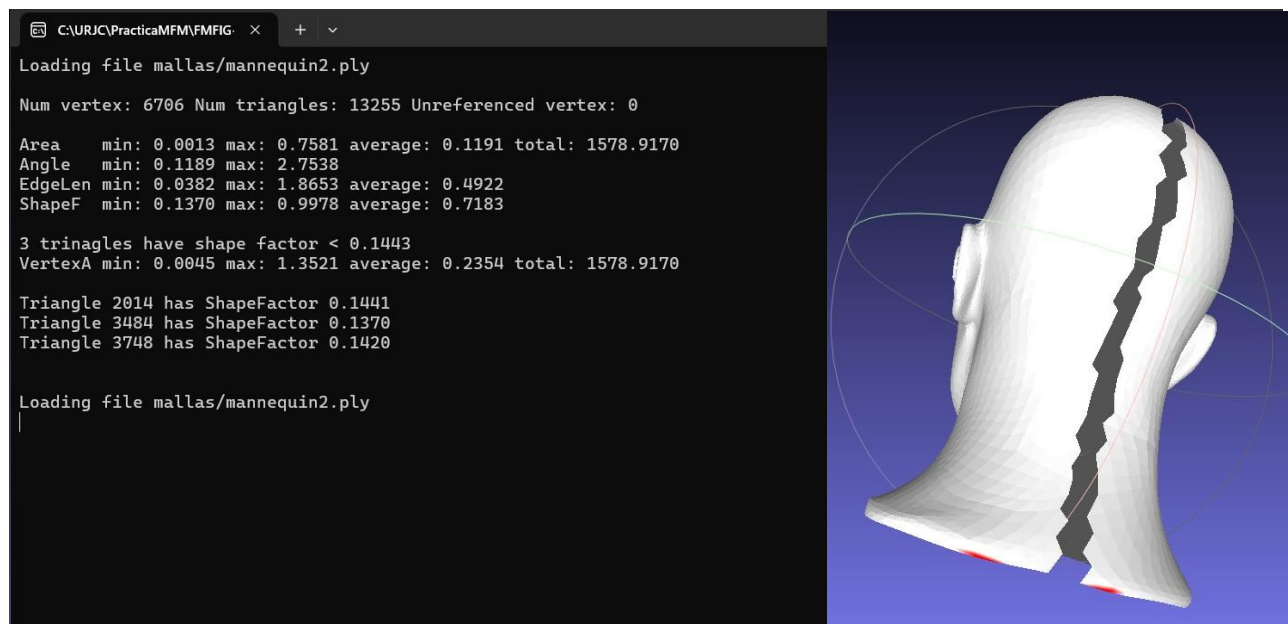
Memoria practica

Estructuras y Algoritmos con Mallas de Triángulos

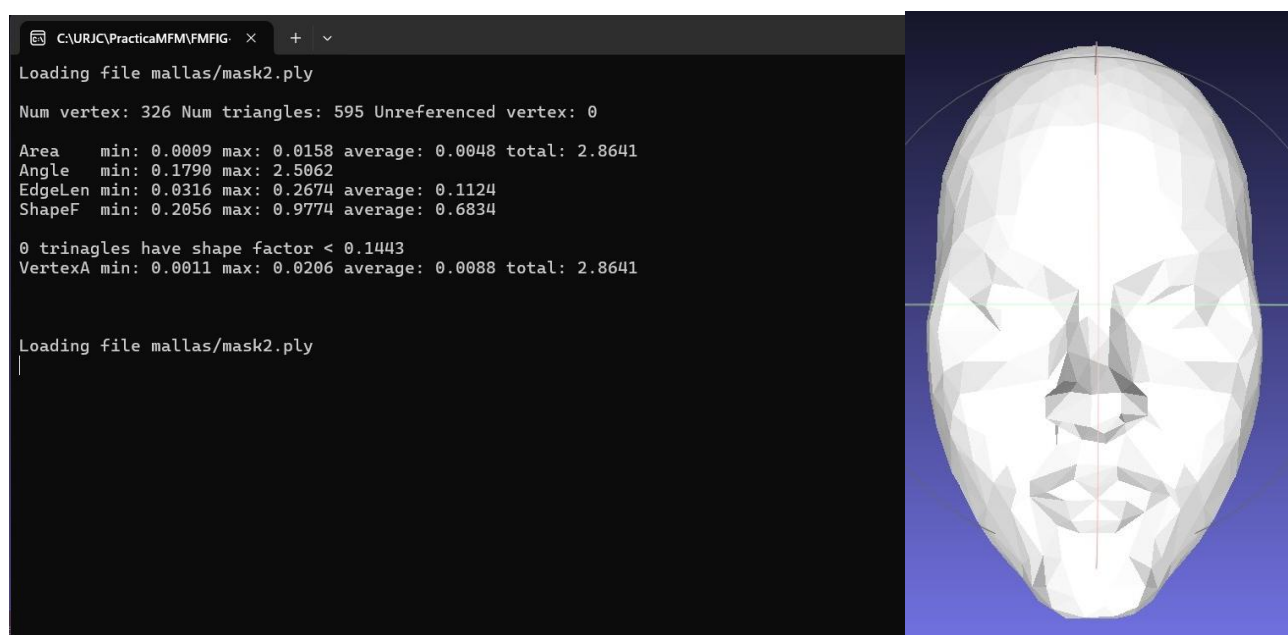
Ejercicio 1: meshStatistic

Parte obligatoria + parte optativa

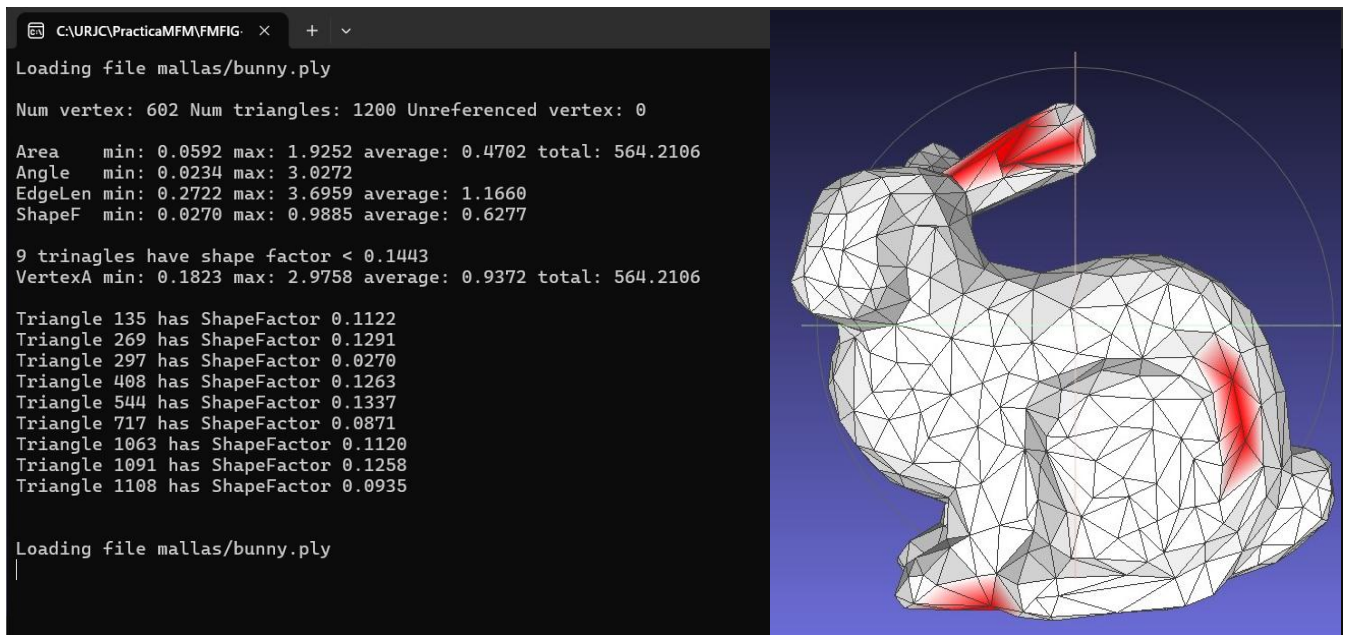
mannequin2



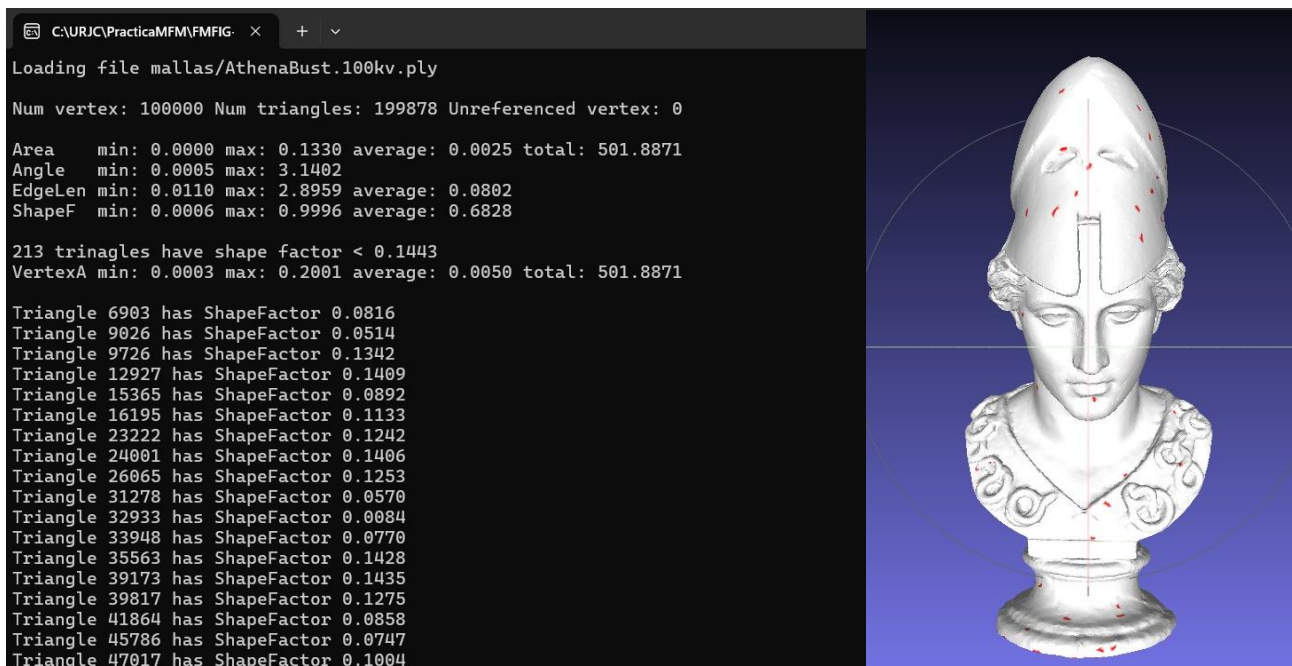
mask2



bunny



AthenaBust.100kv

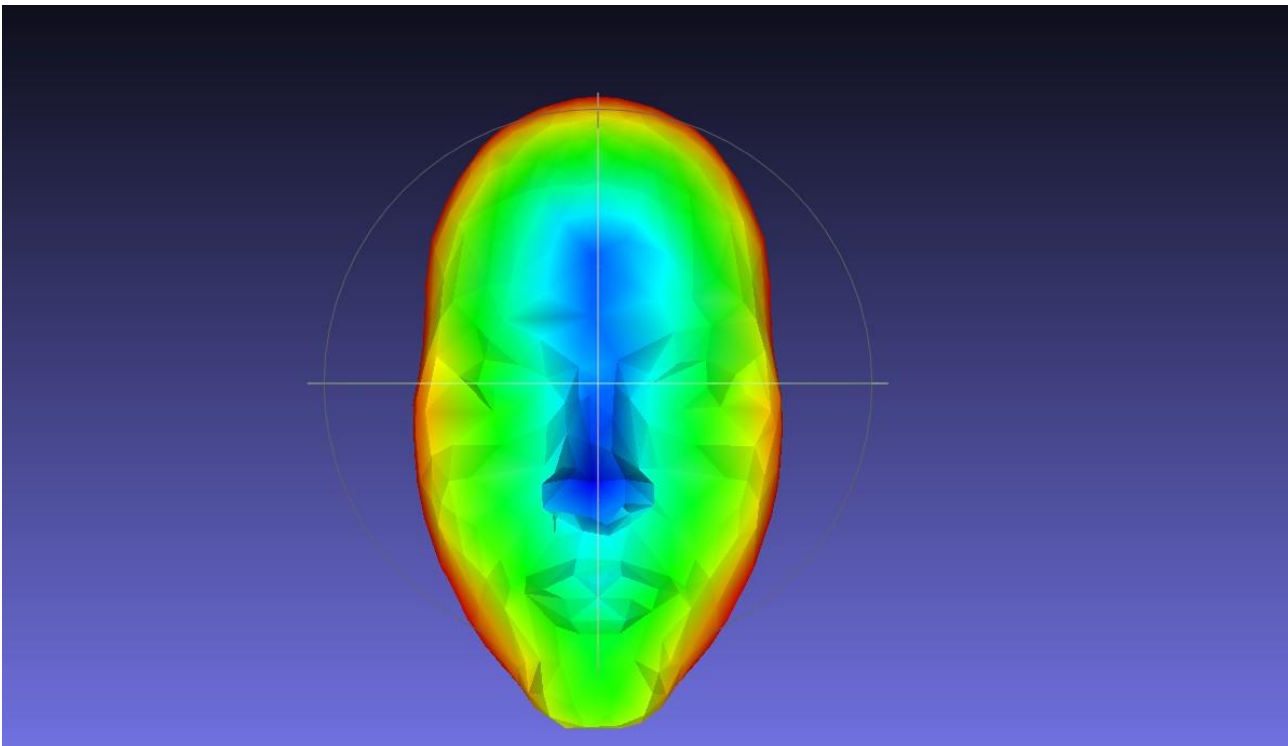


Ejercicio 2: meshBoundary

Parte obligatoria + parte optativa

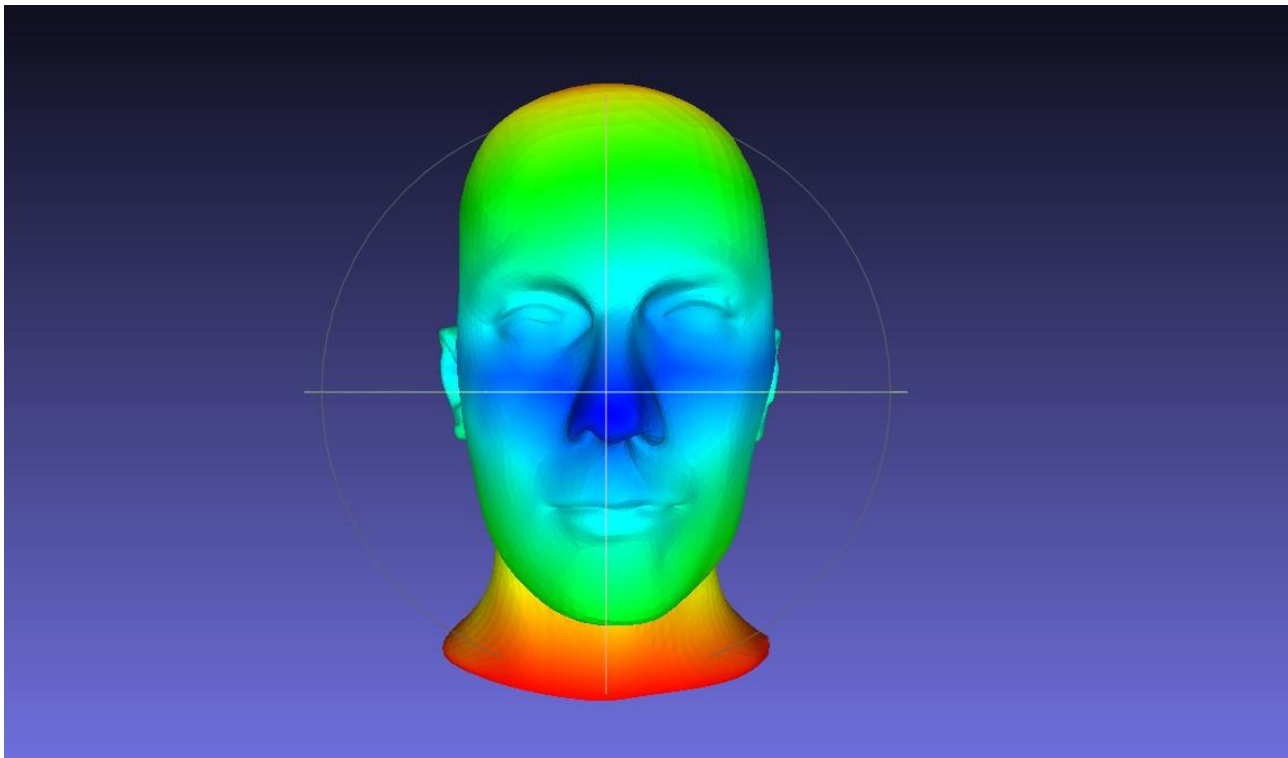
mask2

```
C:\URJC\PracticaMFM\FMFIG: X + v
Loading file mallas/mask2.ply
Num vertex: 326 Num triangles: 595 Unreferenced vertex: 0
Done updateEdgeLists() 0.0002933 seconds
55 boundary edges
865 internal edges
Edges in the boundary:
[27->43] [154->144] [20->13] [16->14] [17->16] [144->143] [13->17] [104->36] [39->104] [105->27] [44->42] [30->44] [42->
45] [123->151] [192->205] [150->165] [151->150] [145->123] [152->154] [45->152] [143->145] [36->105] [198->39] [43->30]
[21->20] [115->21] [114->115] [165->114] [320->319] [172->178] [14->173] [173->175] [175->172] [275->196] [204->187] [20
3->266] [187->263] [317->318] [266->265] [196->313] [301->317] [318->320] [313->316] [315->314] [263->203] [269->189] [3
19->192] [265->198] [205->269] [189->204] [316->315] [314->301] [178->179] [179->276] [276->275]
Boundary length: 5.19081
Euler Characteristic: 1
Done boundDist() 0.0163563 seconds
maxDistance to boundary: 0.866627 at vertex: 99
Saving output to output_boundary.ply
Saving output to C:/URJC/PracticaMFM/Practica_FilippoFalzi/Es2/output_boundary.ply
EXCEPTION: C:/URJC/PracticaMFM/Practica_FilippoFalzi/Es2/output_boundary.ply: Error creating the file
Press Return to end the program
```



mannequin2

```
C:\URJC\PracticaMFM\FMFIG- x + v
Loading file mallas/mannequin2.ply
Num vertex: 6706 Num triangles: 13255 Unreferenced vertex: 0
Done updateEdgeLists() 0.0098273 seconds
155 boundary edges
19805 internal edges
Edges in the boundary:
[70->68] [743->744] [7->8] [11->10] [616->617] [3943->3941] [6->12] [8->11] [384->177] [39->35] [612->619] [18->19] [37->39] [19->1
3] [21->22] [12->14] [43->6] [14->21] [35->18] [36->37] [22->40] [45->43] [41->45] [42->41] [68->42] [69->70] [2208->2206] [617->61
1] [2693->2691] [1500->1501] [10->9] [132->5948] [40->131] [134->132] [178->175] [131->134] [142->144] [143->142] [2689->2545] [394
1->5614] [175->174] [176->183] [183->179] [5337->143] [2467->2359] [174->36] [177->178] [2206->2386] [233->69] [71->5336] [2691->26
89] [750->752] [4838->4905] [179->313] [5247->5248] [313->312] [583->588] [5497->5499] [386->384] [4318->4316] [2389->2693] [4313->
4318] [2362->2365] [588->586] [2461->2462] [611->609] [1506->1502] [619->618] [586->653] [653->612] [5246->743] [312->5796] [749->7
47] [752->749] [1078->616] [2387->2391] [2359->2360] [1079->1078] [1503->1511] [1502->1500] [6174->583] [1505->1506] [1511->1507] [
585->5496] [2205->2204] [6343->7] [2204->2208] [2200->2205] [2360->2362] [2365->2203] [3437->3434] [2386->2387] [2391->2389] [2935-
>2461] [2462->2464] [2464->2467] [1507->1505] [3434->3432] [3949->3946] [2941->2942] [2937->2935] [2942->2937] [3937->3949] [2545->
3440] [3440->3437] [9->71] [144->6174] [5248->5246] [3946->3943] [4311->4313] [4310->4311] [5796->750] [5933->5930] [4905->4906] [4
908->3937] [4906->4908] [3684->5197] [5197->5198] [5200->2941] [5198->5200] [6027->1079] [609->585] [747->5245] [5336->5337] [4316-
>5936] [5496->5497] [5499->233] [1501->5247] [5245->5600] [5598->1503] [5600->5598] [5608->3684] [5611->5608] [5614->5611] [3432->5
667] [5940->5942] [5667->5669] [5672->4310] [5669->5672] [5797->386] [744->5797] [5930->5928] [5936->5933] [618->5939] [5939->5940]
[5942->2200] [5948->176] [2203->5941] [5941->6029] [6029->6027] [13->6343] [6700->4838] [6702->6700] [5928->6704] [6704->6702]
Boundary length: 130.406
Euler Characteristic: 1
Done boundDist() 0.0556979 seconds
maxDistance to boundary: 20.4994 at vertex: 5187
Saving output to output_boundary.ply
Saving output to C:\URJC\PracticaMFM\FMFIG-prac2025/output_boundary.ply
Done colorize by distance to boundary 0.0673401 seconds
Executing external command: "C:/Programmi/VCG/MeshLab/meshlab.exe" C:\URJC\PracticaMFM\FMFIG-prac2025/output_boundary.ply
```



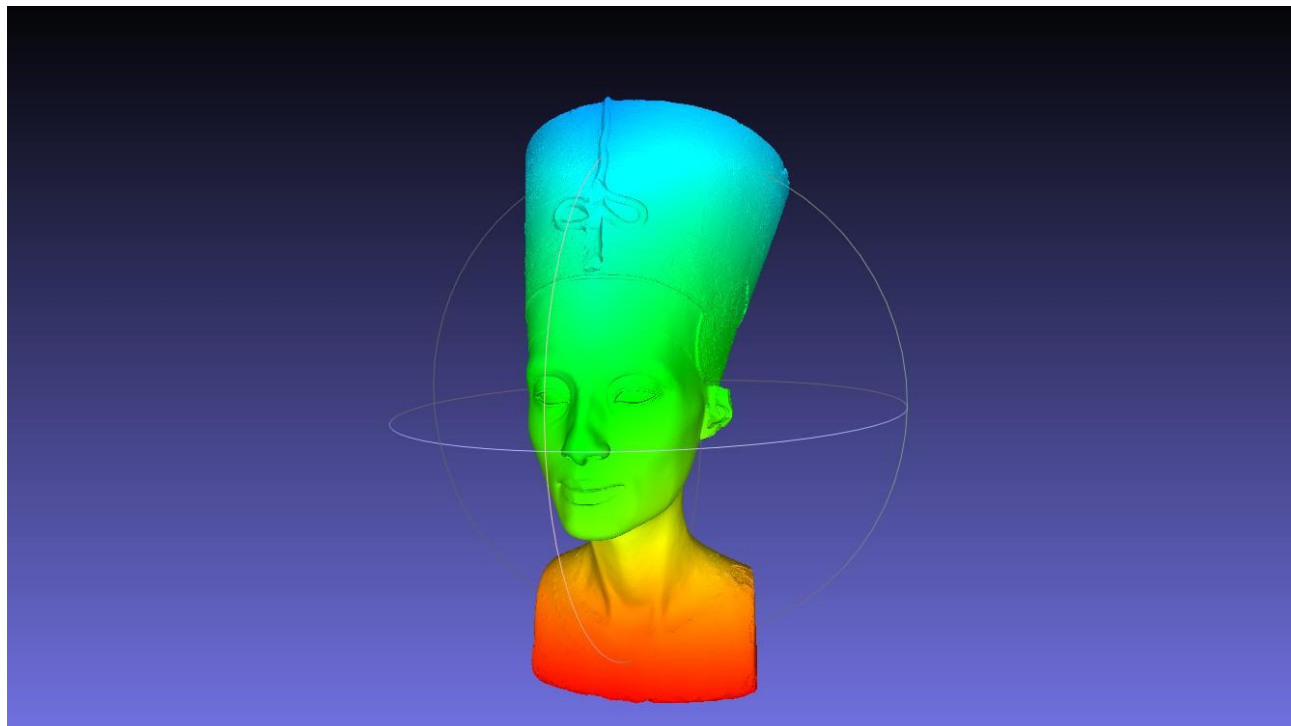
Nefertiti.990kv

```
C:\URJC\PracticaMFM\FMFIG: X + v

Loading file mallas/Nefertiti.990kv.ply
Num vertex: 990781 Num triangles: 1980517 Unreferenced vertex: 0
Done updateEdgeLists() 1.86989 seconds
1043 boundary edges
2970254 internal edges
Edges in the boundary:
[498412->498413] [873700->873701] [914238->914293] [865345->865296] [914945->914939] [866176->866295] [131205->131215] [914325->914
336] [866251->866133] [872543->872542] [131167->131168] [866621->866620] [846400->846401] [131155->131154] [866064->866164] [866564
->866600] [914070->914049] [131159->131117] [913451->913548] [131152->131197] [914307->914309] [131199->131200] [131213->131211] [1
31111->131149] [131204->131181] [382088->390997] [913607->913525] [866290->866291] [866470->866471] [131125->131165] [913499->91349
8] [914659->914639] [131178->131147] [131119->131160] [915719->915717] [382108->382107] [913993->913994] [914499->914419] [131146->
131214] [131197->131196] [866572->866574] [913634->913632] [113736->113731] [914645->914594] [131201->131167] [131164->131135] [131
200->131204] [131216->131203] [131113->131114] [131084->131125] [131149->131084] [131117->131157] [131157->131158] [131160->131161]
[131161->131159] [131163->131123] [131176->131174] [113730->131176] [131179->131180] [131180->131178] [131195->131111] [131114->13
1195] [131196->131113] [131154->131152] [131135->113736] [131165->131164] [131168->131199] [131170->131201] [113731->113730] [13118
1->131179] [131147->131205] [131215->131146] [131214->131213] [874219->873010] [131471->131119] [138302->138303] [873840->873808] [
913973->914046] [390933->390977] [865248->865238] [913233->913234] [913562->913685] [872552->872543] [866527->866513] [865109->8650
89] [921370->921290] [967533->967532] [865232->865221] [913370->913552] [872466->872438] [866279->866280] [967151->967174] [914191->
914192] [866153->866176] [866568->866569] [913386->913297] [921331->921240] [967123->967154] [498441->498439] [872463->872455] [91
3570->913688] [382054->382053] [382062->382079] [382087->382062] [382089->382087] [390960->390979] [382070->382069] [390955->382060
] [390964->390966] [390966->390963] [131198->131155] [382053->390973] [382107->390976] [390976->390975] [390975->382054] [390977->3
82070] [390978->390933] [390949->390978] [382059->390949] [382060->382059] [382071->390964] [390983->390982] [390982->382071] [1311
58->131198] [390992->390991] [390974->131163] [382069->382108] [390993->390955] [390979->390993] [382073->390994] [390995->382073]
[131123->390992] [390997->390995] [382079->382088] [390973->390974] [390991->131471] [866618->866622] [872448->846381] [390963->390
960] [390994->390983] [872551->872979] [866287->866288] [866579->866525] [498414->498398] [498413->498414] [498410->498412] [498420
->498410] [498421->498420] [498418->498421] [498425->498426] [498426->498418] [498431->498429] [498439->498431] [498444->498441] [4
98481->498482] [921468->921289] [872562->872563] [921039->921112] [912443->912410] [866601->866603] [865084->865077] [913913->91398
3] [913706->913645] [921046->921191] [920823->138514] [866577->866523] [846412->846397] [866600->866565] [866489->866358] [914782->
914758] [865221->865210] [865415->865376] [914253->914252] [780482->780483] [780404->780432] [780432->780383] [873016->873021] [913
764->913772] [138514->138405] [913486->913386] [914254->914118] [967229->967228] [914252->914190] [913656->913579] [872469->873730]
[873698->873723] [866538->866423] [920667->920550] [865346->865345] [913601->913672] [914639->914609] [866543->866544] [866578->86
...
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...

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>990423] [990686->990416] [990204->990685] [990695->990738] [990751->990750] [990351->990350] [990350->990737] [990476->990477] [99
0605->990611] [990310->990309] [990740->990712] [990657->990656] [990656->990593] [990737->990657] [990595->990767] [990750->990699
] [990702->990751] [990477->990647] [990778->990757] [990712->990759] [990759->990479] [990479->990713] [990475->990740] [990757->9
90760] [990713->990700] [990760->990780] [990780->990475] [990773->990779] [990779->990778] [990647->990773] [990767->990696] [9901
83->990182] [990223->990221] [990028->990351]
Boundary length: 644.293
Euler Characteristic: 1
Done boundDist() 2.82965 seconds
maxDistance to boundary: 628.778 at vertex: 429856
Saving output to output_boundary.ply
Saving output to C:/URJC/PracticaMFM/FMFIG-prac2025/output_boundary.ply
Done colorize by distance to boundary 4.10489 seconds
Executing external command: "C:/Programmi/VCG/MeshLab/meshlab.exe" C:/URJC/PracticaMFM/FMFIG-prac2025/output_boundary.ply
```

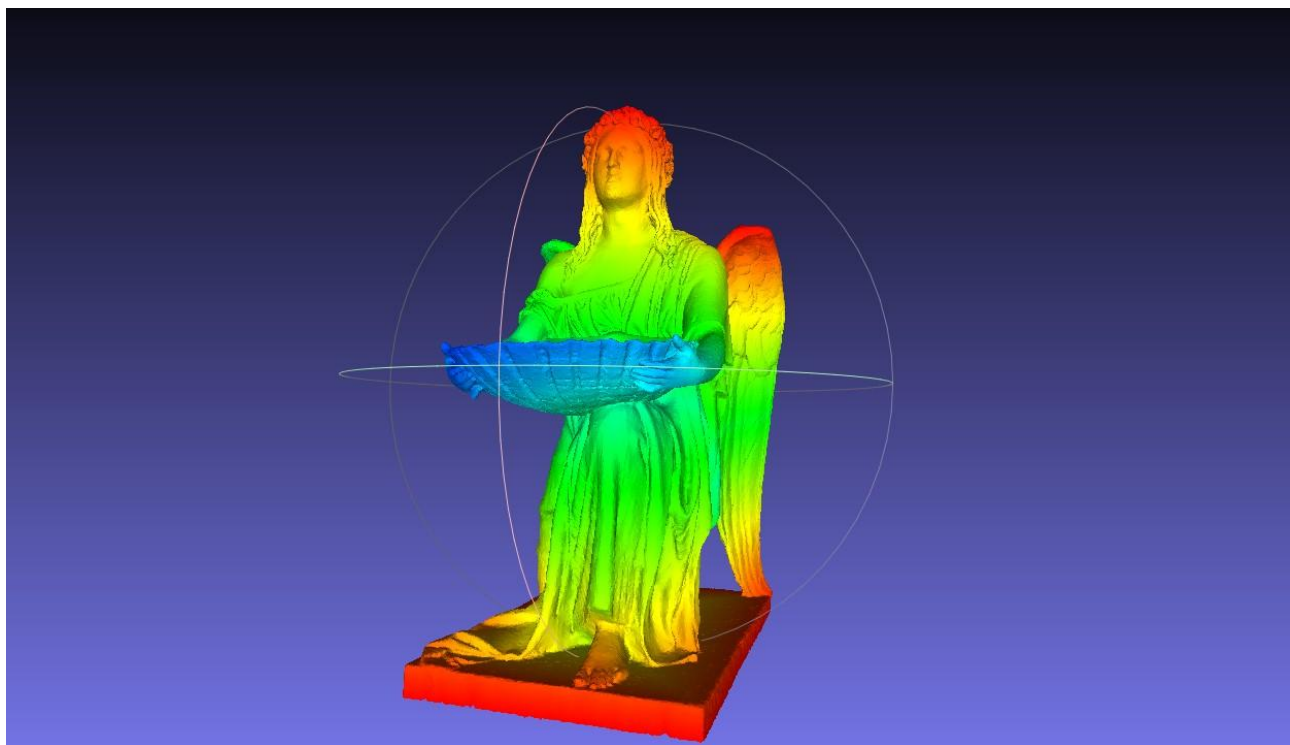


angel_kneeling

```
C:\URJC\PracticaMFM\FMFIG x + v
Loading file mallas\angel_kneeling.150kv.ply
Num vertex: 150372 Num triangles: 299992 Unreferenced vertex: 0
Done updateEdgeLists() 0.215597 seconds
766 boundary edges
449605 internal edges
Edges in the boundary:
[85715->91391] [5808->5805] [52750->52529] [102079->102349] [3739->3478] [3740->3393] [3333->3039] [25492->48180] [3393->3332] [3659->3633] [3518->3611] [75
000->75163] [3602->3722] [35997->25490] [75578->75579] [3229->3735] [26759->26744] [86834->86836] [3633->3719] [112384->112343] [3530->3529] [3736->3333] [3
737->3663] [3443->3650] [83861->82310] [3611->3659] [83259->85604] [3529->3740] [3708->3530] [3515->3064] [72934->75675] [3131->3557] [26540->26556] [3332->
3736] [3478->3079] [3187->3186] [3712->3687] [3079->3704] [3663->3235] [3719->3331] [3728->3191] [3177->3708] [3350->3187] [5809->3728] [3735->3161] [3126->
3500] [3353->3515] [3723->3680] [3722->3350] [52692->52741] [5805->5809] [3161->3518] [3557->3712] [3140->3177] [74513->75208] [3235->3723] [3186->3131] [36
84->3353] [3579->3444] [3687->3739] [112489->112480] [3444->3443] [3191->3140] [3680->3579] [3650->3694] [3704->3229] [3064->3602] [3039->3126] [3500->3737]
[3331->75273] [26463->35998] [3694->3684] [90572->85649] [75734->75791] [26441->32246] [85733->85724] [86861->86746] [75175->75773] [89476->91803] [112256-
>112218] [75625->75424] [91737->91549] [75799->75287] [102930->103348] [49530->49466] [74308->72797] [102136->102018] [5811->5800] [5800->5808] [4860->5811]
[91803->85715] [89550->91796] [75579->75625] [52751->52723] [74768->74394] [102078->101725] [102258->102337] [102359->101876] [26586->26518] [90176->91772]
[74987->74988] [32272->35683] [112527->112497] [29188->29223] [74459->74191] [111748->111776] [101725->102332] [49436->49465] [50934->50972] [103268->10308
6] [85646->91823] [91693->91819] [74114->74768] [75441->75045] [133121->132416] [84539->85642] [26558->26537] [72830->75533] [85772->85725] [75130->74987] [
91339->90757] [102832->103197] [75314->74513] [29172->29224] [26506->26519] [86780->86779] [51051->51054] [85403->84539] [85504->83260] [32289->32272] [5252
9->52677] [85658->90176] [75533->72639] [90736->85646] [48865->48859] [102452->103046] [75332->75333] [103293->103323] [26770->26585] [25490->25492] [112515
```

...

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[132783->132936] [132761->132532] [132708->132928] [133038->132976] [133095->132462] [132749->133095] [133030->100064] [132462->133038] [133117->133077] [
132374->132685] [133041->132760] [132634->132869] [133033->132884] [91081->133058] [132923->133024] [132632->132631] [132809->133113] [132384->133075] [1331
09->133053] [132416->132374] [132884->132684] [133091->133111] [133119->133020] [132631->133119] [133102->132919] [132518->133074] [133105->133041] [133115-
>133098] [133071->132947] [132976->132540] [142141->142279] [142234->142141] [142156->141683] [142279->142244] [141683->142234] [143245->142595] [142490->14
3551] [143425->143696] [143551->143575] [143636->142807] [143624->143520] [143722->143245] [142595->143014] [143575->143425] [143520->143722] [142244->14362
4] [142807->142459] [142459->138934] [143696->143623] [143014->142490] [149579->142156] [150239->147467] [148898->147120] [149795->148898] [150362->150323] [
147467->150357] [149721->149746] [150306->147282] [149582->149579] [147120->150362] [147303->149795] [149746->150260] [147282->150370] [150370->150239] [15
0323->150224] [150224->150317] [150317->150306] [138934->149721] [150357->149582]
Boundary length: 50.2121
Euler Characteristic: -7
Done boundDist() 0.580894 seconds
maxDistance to boundary: 13.7901 at vertex: 44431
Saving output to output_boundary.ply
Saving output to C:/URJC/PracticaMFM/FMFIG-prac2025/output_boundary.ply
Done colorize by distance to boundary 0.776607 seconds
Executing external command: "C:/Programmi/VCG/MeshLab/meshlab.exe" C:/URJC/PracticaMFM/FMFIG-prac2025/output_boundary.ply
```

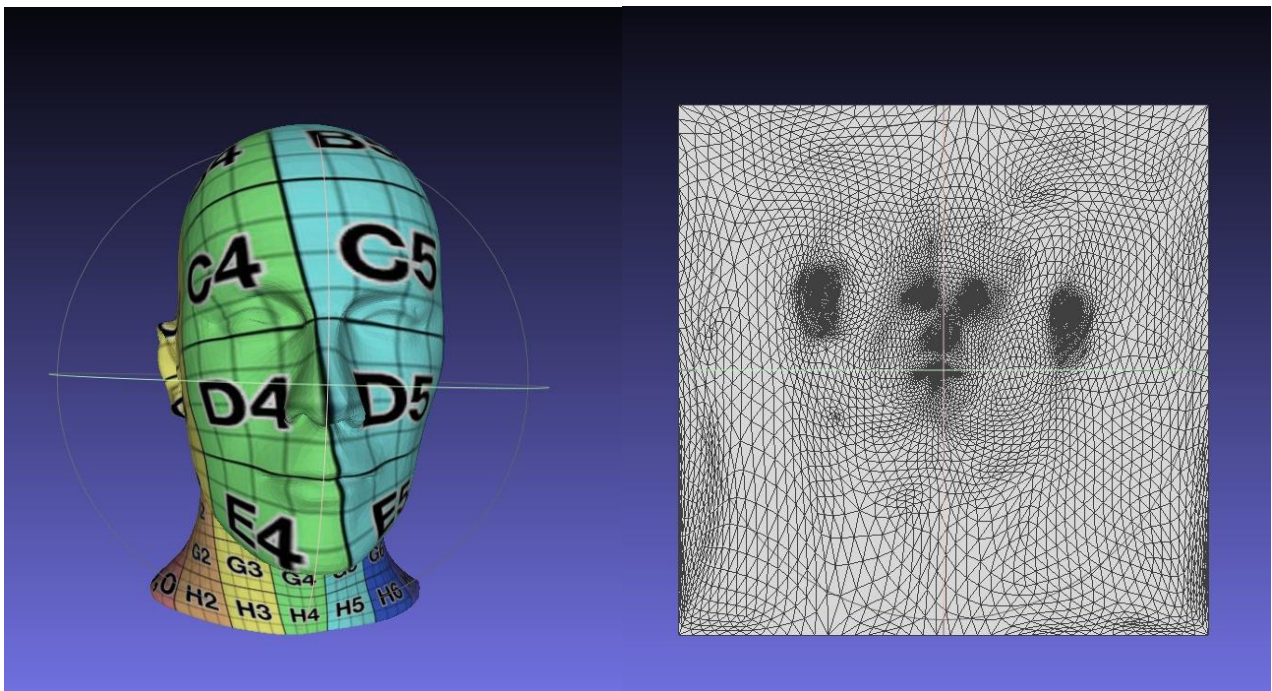


Ejercicio 3: parte obligatoria + parte optativa

Parte obligatoria: meshTexture-Dense

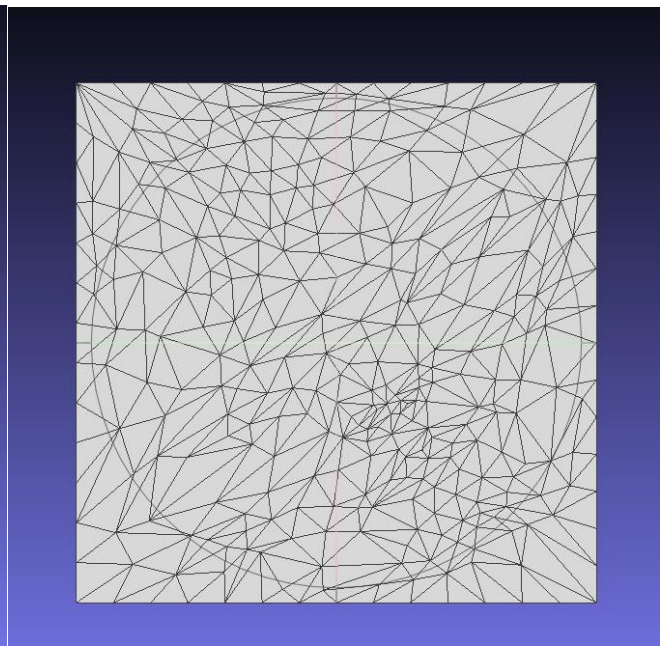
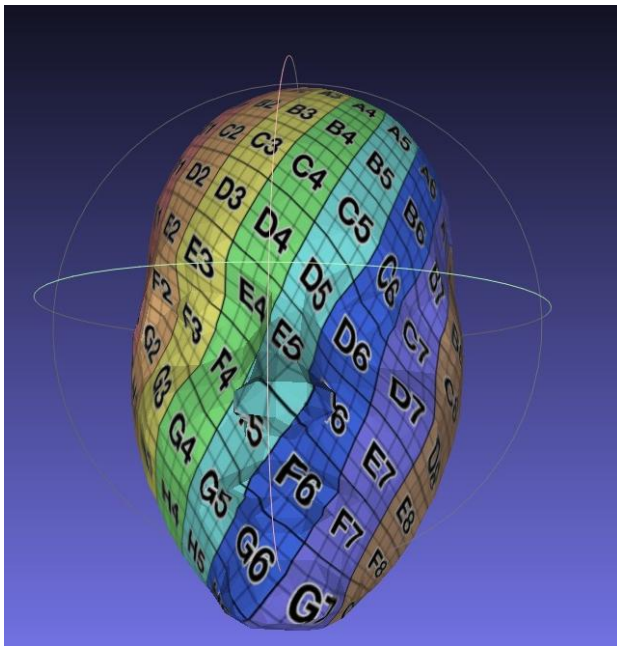
mannequin2

```
C:\URJC\PracticaMFM\FMFIG x + v
Loading file mallas/mannequin2.ply
Num vertex: 6706 Num triangles: 13255 Unreferenced vertex: 0
155 vertex in external boundary:
Done compute Boundary. 0.0084271 seconds
Computing Laplacian matrix (dense):
Done Laplacian matrix (dense). 0.0974947 seconds
Computing System matrix:
Done System matrix. 0.110622 seconds
Computing contour conditions:
Done contour conditions. 0.111596 seconds
Solving the system using Eigen (dense):
Done solving the system. 34.9931 seconds
Saving parameterization mesh to output_UVMesh1.ply
Saving texture mesh to output_UVMesh2.ply
Executing external command: "C:/Programmi/VCG/MeshLab/meshlab.exe" output_UVMesh2.ply
```



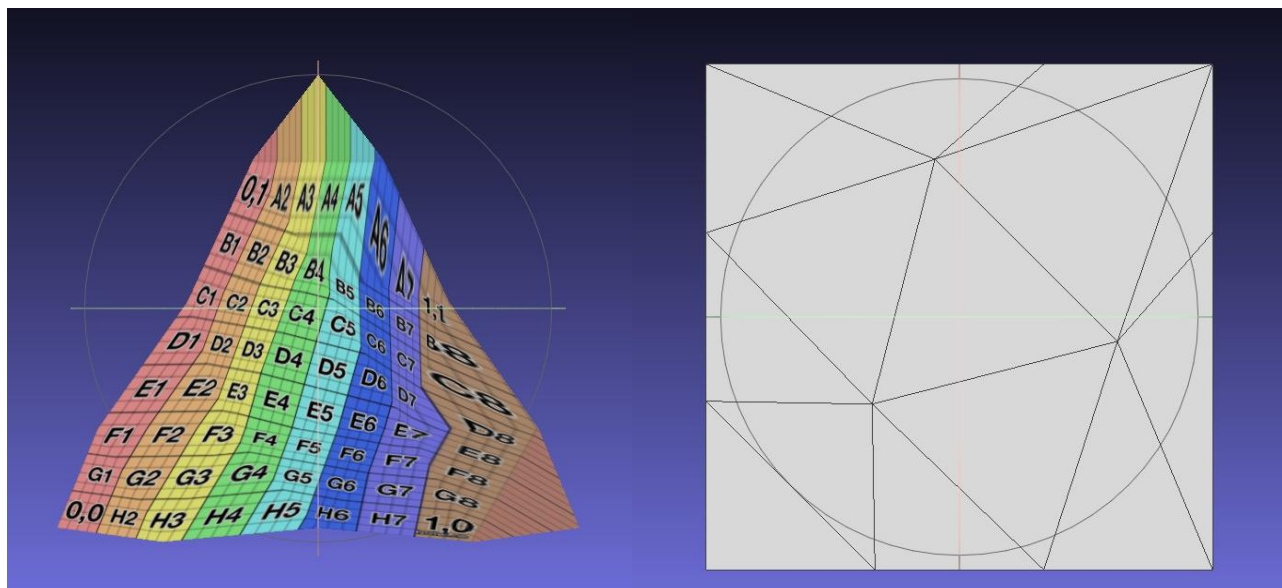
mask2

```
C:\URJC\PracticaMFM\FMFIG- x + v
Loading file mallas/mask2.ply
Num vertex: 326 Num triangles: 595 Unreferenced vertex: 0
55 vertex in external boundary:
[27->43] [20->13] [16->14] [17->16] [13->17] [30->44] [114->115] [104->36] [39->104] [105->27] [44->42] [42->45] [123->1
51] [150->165] [144->143] [318->320] [151->150] [45->152] [145->123] [152->154] [154->144] [143->145] [36->105] [198->39
] [43->30] [21->20] [115->21] [165->114] [204->187] [172->178] [14->173] [173->175] [175->172] [275->196] [203->266] [18
7->263] [192->205] [317->318] [196->313] [301->317] [313->316] [315->314] [320->319] [263->203] [269->189] [319->192] [2
65->198] [266->265] [205->269] [189->204] [316->315] [314->301] [178->179] [179->276] [276->275]
Done compute Boundary. 0.0143072 seconds
Computing Laplacian matrix (dense):
Done Laplacian matrix (dense). 0.0154436 seconds
Computing System matrix:
Done System matrix. 0.0162061 seconds
Computing contour conditions:
Done contour conditions. 0.0168174 seconds
Solving the system using Eigen (dense):
Done solving the system. 0.0251948 seconds
Saving parameterization mesh to output_UVMesh1.ply
Saving texture mesh to output_UVMesh2.ply
Executing external command: "C:/Programmi/VCG/MeshLab/meshlab.exe" output_UVMesh2.ply
```



16Triangles

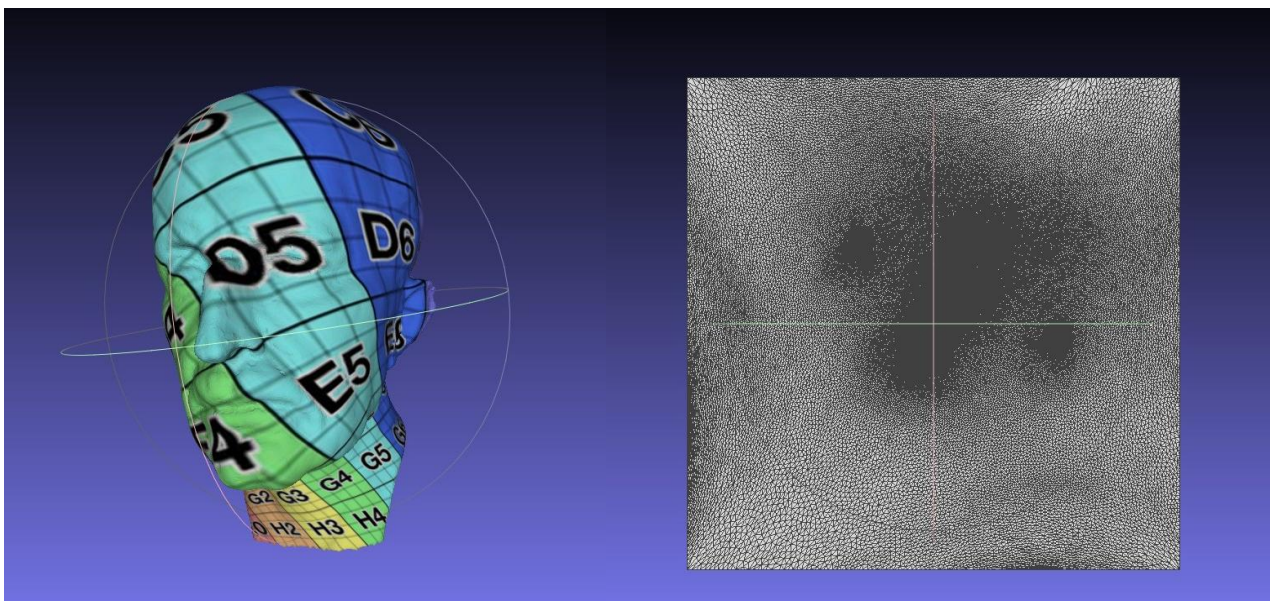
```
C:\URJC\PracticaMFM\FMFIG: x + v
Loading file mallas/16Triangles.off
Num vertex: 15 Num triangles: 16 Unreferenced vertex: 0
12 vertex in external boundary:
[10->3] [9->1] [2->9] [1->10] [12->2] [0->11] [11->4] [14->0] [4->12] [5->14] [3->13] [13->5]
Done compute Boundary. 0.0039787 seconds
Computing Laplacian matrix (dense):
Done Laplacian matrix (dense). 0.0047087 seconds
Export matrix L to file: laplacian.mat
To import L into matlab use the command: load("laplacian.mat")
Computing System matrix:
Done System matrix. 0.007058 seconds
Export matrix A to file: systemMatrix.mat
To import A into matlab use the command: load("systemMatrix.mat")
Computing contour conditions:
Done contour conditions. 0.0097921 seconds
Export matrix UV0 to file: boundaryUV0.mat
To import UV0 into matlab use the command: load("boundaryUV0.mat")
Solving the system using Eigen (dense):
Done solving the system. 0.0117629 seconds
Export matrix UV to file: solutionUV.mat
To import UV into matlab use the command: load("solutionUV.mat")
Saving parameterization mesh to output_UVMesh1.ply
Saving texture mesh to output_UVMesh2.ply
Executing external command: "C:/Programmi/VCG/MeshLab/meshlab.exe" output_UVMesh2.ply
```



Parte optativa: meshTexture-Sparse

maxplanck.45kv

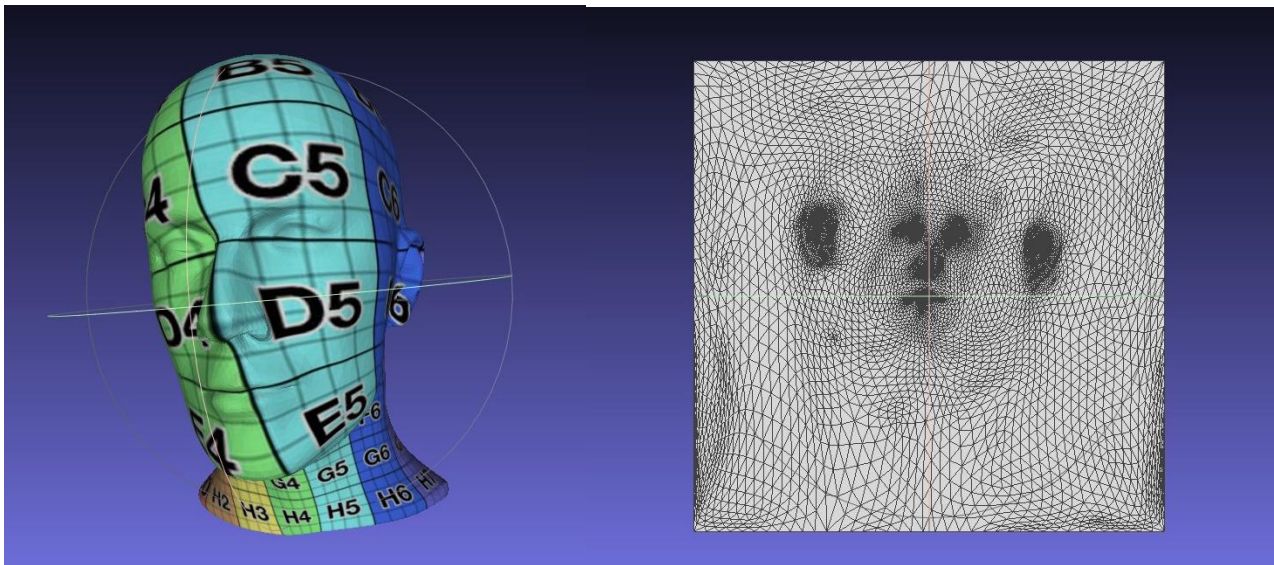
```
C:\URJC\PracticaMFM\FMFIG- x + v
Loading file mallas/maxplanck.45kv.ply
Num vertex: 45045 Num triangles: 89641 Unreferenced vertex: 0
447 vertex in external boundary:
Done compute Boundary. 0.0571939 seconds
Computing Laplacian matrix (sparse):
Done Laplacian matrix (sparse). 0.074003 seconds
Computing System matrix:
Done System matrix. 0.101677 seconds
Computing contour conditions:
Done contour conditions. 0.103197 seconds
Solving the system using Eigen (sparse):
Done solving the system. 0.607095 seconds
Saving parameterization mesh to output_UVMesh1.ply
Saving texture mesh to output_UVMesh2.ply
Executing external command: "C:/Programmi/VCG/MeshLab/meshlab.exe" output_UVMesh2.ply
```



mannequin2

```
Console di debug di Microsoft
Loading file mallas/mannequin2.ply
Num vertex: 6706 Num triangles: 13255 Unreferenced vertex: 0
155 vertex in external boundary:
Done compute Boundary. 0.0066286 seconds
Computing Laplacian matrix (sparse):
Done Laplacian matrix (sparse). 0.0099715 seconds
Computing System matrix:
Done System matrix. 0.0154224 seconds
Computing contour conditions:
Done contour conditions. 0.0161893 seconds
Solving the system using Eigen (sparse):
Done solving the system. 0.0570675 seconds
Saving parameterization mesh to output_UVMesh1.ply
Saving texture mesh to output_UVMesh2.ply
Executing external command: "C:/Programmi/VCG/MeshLab/meshLab.exe" output_UVMesh2.ply

C:\URJC\PracticaMFM\Mallas\FMFIG-prac2025\x64\Release\meshTexture-Sparse.exe (processo 31244) terminato. Codice restituito: 0 (0x0).
Premere un tasto qualsiasi per chiudere questa finestra...|
```



laurana50k

```
C:\URJC\PracticaMFM\FMFIG. x + v
Loading file mallas/laurana50k.ply
Num vertex: 25030 Num triangles: 49768 Unreferenced vertex: 0
290 vertex in external boundary:
Done compute Boundary. 0.0405846 seconds
Computing Laplacian matrix (sparse):
Done Laplacian matrix (sparse). 0.0530379 seconds
Computing System matrix:
Done System matrix. 0.0724246 seconds
Computing contour conditions:
Done contour conditions. 0.074087 seconds
Solving the system using Eigen (sparse):
Done solving the system. 0.300835 seconds
Saving parameterization mesh to output_UVMesh1.ply
Saving texture mesh to output_UVMesh2.ply
Executing external command: "C:/Programmi/VCG/MeshLab/meshlab.exe" output_UVMesh2.ply
|
```

